

INTRODUCTION TO APPLIED NUMERICAL METHODS

LU Decomposition

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LU DECOMPOSITION(CONT)

The system, $[A]\{x\}=\{b\}$, can be decomposed into two matrices

$[L]$: lower triangular matrix

$[U]$: upper triangular matrix

such that

$$\begin{aligned}[L][U]&=[A] \\ [L][U]\{x\}&=\{b\}\end{aligned}$$

Set up the following systems

$$\begin{aligned}[L]\{d\}&=\{b\} \\ [U]\{x\}&=\{d\}\end{aligned}$$

Step 1: $[L]\{d\}=\{b\}$ is used to generate an intermediate vector $\{d\}$ by forward substitution.

Step 2: $[U]\{x\}=\{d\}$ is used to get the answer, $\{x\}$, by back substitution.

LU DECOMPOSITION-BASED SOLUTION PROCESS

Process

- **Step 1:** LU Decomposition: $[A]$ is factorized or “decomposed” into lower $[L]$ and upper $[U]$ triangular matrices, i.e., $[A] = [L][U]$
- **Step 2:** Substitution step: (i)

$$[L][U]\underbrace{\{x\}}_{\{D\}} = \{B\}$$

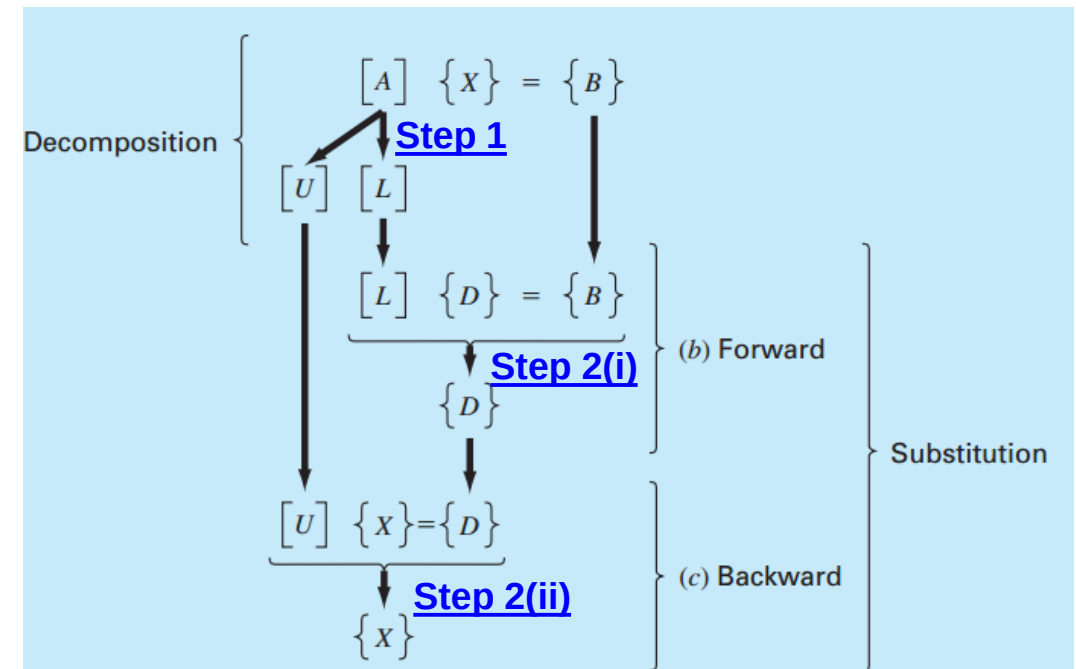
Introduce an intermediate vector $\{D\}$

$$[L]\{D\} = \{B\}$$

Forward Substitution

$$\{D\}$$

Backward Substitution (ii) $[U]\{X\} = \{D\}$



LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

[L] and [U] is produced from the row operation during forward elimination step

Step 1: Forward Elimination

Matrix/Vector Form

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix}$$

$$\downarrow [A]: \text{Row}(2) - \text{Row}(1) \times \left(\frac{a_{21}}{a_{11}} \right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{Define: } f_{21} = \left(\frac{a_{21}}{a_{11}} \right)$$

$$\downarrow [A]: \text{Row}(3) - \text{Row}(1) \times \left(\frac{a_{31}}{a_{11}} \right)$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$

$$\text{Define: } f_{31} = \left(\frac{a_{31}}{a_{11}} \right)$$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

[L] and [U] is produced from the row operation during forward elimination step



Matrix/Vector Form

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & a'_{32} & a'_{33} \end{bmatrix}$$



$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

$$[A]: \text{Row}(3') - \text{Row}(2') \times \left(\frac{a'_{32}}{a'_{22}} \right)$$

Define: $f_{32} = \left(\frac{a'_{32}}{a'_{22}} \right)$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

Key idea: use f_{21} , f_{31} , f_{32} , ... to assemble the lower triangular matrix $[L]$

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix}$$

$$[U] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix} \quad \rightarrow \quad [A] \rightarrow [L][U]$$

$$\begin{aligned} l_{11} &= 1 & l_{12} &= 0 & l_{13} &= 0 \\ l_{21} &= f_{21} & l_{22} &= 1 & l_{23} &= 0 \\ l_{31} &= f_{31} & l_{32} &= f_{32} & l_{33} &= 1 \end{aligned}$$

$$\begin{aligned} u_{11} &= a_{11} & u_{12} &= a_{12} & u_{13} &= a_{13} \\ u_{21} &= 0 & u_{22} &= a'_{22} & u_{23} &= a'_{23} \\ u_{31} &= 0 & u_{32} &= 0 & u_{33} &= a''_{33} \end{aligned}$$

LU DECOMPOSITION VERSION OF GAUSS ELIMINATION

Process

- **Step 2(i):** Forward substitution step to obtain $\{D\}$

$$[L]\{D\} = \{B\}$$

$$d_i = b_i - \sum_{j=1}^{i-1} l_{ij} d_j \quad \text{for } i = 2, 3, \dots, n$$

$$\begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

- **Step 2(ii):** Back-substitution step to obtain $\{x\}$

$$[U]\{X\} - \{D\} = 0$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix}$$

$$x_n = \frac{d_n}{u_{nn}}$$

$$x_i = \frac{d_i - \sum_{j=i+1}^n u_{ij} d_j}{u_{ii}} \quad \text{for } i = n-1, n-2, \dots, 1$$

AN EXAMPLE

Derive an LU decomposition based on the Gauss elimination using the previous example

Find the solution to x based on LU decomposition

Solution: The matrix form of the equation set

$$\begin{aligned} 3x_1 - 0.1x_2 - 0.2x_3 &= 7.85 \\ 0.1x_1 + 7x_2 - 0.3x_3 &= -19.3 \\ 0.3x_1 - 0.2x_2 + 10x_3 &= 71.4 \end{aligned}$$

$$[A] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.1 & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix}$$

Following forward elimination, the upper triangular matrix $[U]$ was

$$[U] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix}$$

Determine the lower triangular matrix $[L]$: need factors for forward elimination ([see next several slides](#))

$$f_{21} = \frac{0.1}{3} = 0.03333333 \quad f_{31} = \frac{0.3}{3} = 0.1000000 \quad f_{32} = \frac{-0.19}{7.00333} = -0.0271300$$

AN EXAMPLE: FORWARD ELIMINATION

Use Gauss elimination to solve

$$[A] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.1 & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix} \quad \begin{matrix} (1) \\ (2) \\ (3) \end{matrix}$$

Solution: Forward Elimination

Step 1.1:

Row (1) $\times$ (0.1/3)

$$\begin{bmatrix} 0.1 & -\frac{0.01}{3} & -\frac{0.02}{3} \end{bmatrix} \quad (1^*)$$

$$\textcircled{1} \times \left(\frac{0.1}{3} \right)$$

$$f_{21} = \left(\frac{0.1}{3} \right)$$

Row (2) – Row (1*)

$$\begin{bmatrix} 0.1 & 7 & -0.3 \end{bmatrix} \quad (2)$$

$$\begin{array}{r} \begin{bmatrix} 0.1 & -\frac{0.01}{3} & -\frac{0.02}{3} \end{bmatrix} \\ - \\ \begin{bmatrix} 0.1 & 7 & -0.3 \end{bmatrix} \end{array} \quad (1^*)$$

New A Matrix

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0.3 & -0.2 & 10 \end{bmatrix}$$

$$\begin{array}{c} \downarrow \\ \begin{bmatrix} 0 & 7.00333 & -0.293333 \end{bmatrix} \end{array}$$

(2') $\rightarrow$

AN EXAMPLE: FORWARD ELIMINATION

A matrix

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0.3 & -0.2 & 10 \end{bmatrix} \quad \begin{matrix} (1) \\ (2') \\ (3) \end{matrix}$$

Solution: Forward Elimination

Step 1.2: Row (1) $\times (0.3/3)$ $\textcircled{1} \times \left(\frac{0.3}{3}\right)$

$$\begin{bmatrix} 0.3 & -0.01 & -0.02 \end{bmatrix} \quad (1^+)$$

$$f_{31} = \left(\frac{0.3}{3}\right)$$

Row (3) – Row (1+)

$$\begin{bmatrix} 0.3 & -0.2 & 10 \end{bmatrix} \quad (3)$$

–

$$\begin{bmatrix} 0.3 & -0.01 & -0.02 \end{bmatrix} \quad (1^+)$$

↓

$$\begin{bmatrix} 0 & -0.19 & 10.02 \end{bmatrix} \quad (3') \rightarrow$$

New A Matrix

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & -0.19 & 10.02 \end{bmatrix} \quad \begin{matrix} (1) \\ (2') \\ (3') \end{matrix}$$

AN EXAMPLE: FORWARD ELIMINATION

$$\begin{array}{ccc} \text{A Matrix} & & \\ \left[\begin{array}{ccc} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & -0.19 & 10.02 \end{array} \right] & \begin{array}{l} (1) \\ (2') \\ (3') \end{array} \end{array}$$

Solution: Forward Elimination

Step 2:

$$\text{Row (2')} \times (-0.19/7.00333) \quad \textcircled{2} \times \left(-\frac{0.19}{7.00333} \right) \quad f_{32} = \left(-\frac{0.19}{7.00333} \right)$$

$$\left[\begin{array}{ccc} 0 & -0.19 & 0.0079581 \end{array} \right] \quad (2^*)$$

Row (3') - Row (2')

$$\left[\begin{array}{ccc} 0 & -0.19 & 10.02 \end{array} \right] \quad (3')$$

—

$$\left[\begin{array}{ccc} 0 & -0.19 & 0.0079581 \end{array} \right] \quad (2^*)$$



$$\left[\begin{array}{ccc} 0 & 0 & 10.012 \end{array} \right] \quad (3'')$$

New A Matrix

$$\left[\begin{array}{ccc} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.012 \end{array} \right]$$

AN EXAMPLE

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4$$

Solution:

Step 3: Determine the lower triangular matrix [L]: need factors for forward elimination

$$f_{21} = \frac{0.1}{3} = 0.03333333 \quad f_{31} = \frac{0.3}{3} = 0.10000000 \quad f_{32} = \frac{-0.19}{7.00333} = -0.0271300$$

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix}$$

The LU decomposition is:

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix}$$

AN EXAMPLE

Solution:

Step 4: $[A] = [L][U]$????

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix}$$

↓ verify

$$[L][U] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.0999999 & 7 & -0.3 \\ 0.3 & -0.2 & 9.99996 \end{bmatrix}$$

AN EXAMPLE

Solution:

Step 5: A forward-substitution to compute $\{D\}$:

$$[L]\{D\} = \{B\}$$

$$d_i = b_i - \sum_{j=1}^{i-1} L_{ij} d_j \quad \text{for } i = 2, 3, \dots, n$$

Forward
substitution
↓

$$\begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.3 \\ 71.4 \end{Bmatrix}$$

$$d_1 = 7.85$$

$$d_2 = -19.3 - 0.0333333(7.85) = -19.5617$$

$$d_3 = 71.4 - 0.1(7.85) + 0.02713(-19.5617) = 70.0843$$

$$\{D\} = \begin{Bmatrix} 7.85 \\ -19.5617 \\ 70.0843 \end{Bmatrix}$$

AN EXAMPLE

Solution:

Step 6: A backward-substitution to compute $\{X\}$: $[U]\{X\} = \{D\}$

$$x_n = \frac{d_n}{u_{nn}}, \quad x_i = \frac{d_i - \sum_{j=i+1}^n u_{ij}d_j}{u_{ii}} \quad \text{for } i = n-1, n-2, \dots, 1$$

Back-substitution ↑

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 7.85 \\ -19.5617 \\ 70.0843 \end{Bmatrix}$$

$$\{X\} = \begin{Bmatrix} 3 \\ -2.5 \\ 7.00003 \end{Bmatrix}$$

MATRIX INVERSION

- Inversion of $[A]$: $[A][A]^{-1} = [I]$
- The inversion can be calculated in a column-by-column fashion by generating solutions with unit vectors as the right-hand-side constants.
- Example:
 - $[A] = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$
 - $[A]^{-1} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$
 - $[A][A]^{-1} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 - Calculated the matrix in a column-by-column fashion (solving two linear equations):
 - $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 - $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_{12} \\ x_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

AN EXAMPLE FOR MATRIX INVERSION

- Employ LU decomposition to determine the matrix inverse for the previous example

$$[A] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.1 & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix}$$

Recall that the decomposition resulted in the following lower and upper triangular matrices:

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix}$$

AN EXAMPLE FOR MATRIX INVERSION

- Solution
 - Calculated the matrix in a column-by-column fashion (solving two linear equations):

$$\begin{aligned} [L][U]\{X_1\} &= \{I_1\} \\ [L][U]\{X_2\} &= \{I_2\} \\ &\dots \\ [L][U]\{X_n\} &= \{I_n\} \end{aligned}$$

Step 1: A forward-substitution to compute {D}: $[L]\{D\} = \{I_1\}$

Forward
substitution ↓

$$[L][U]\{X_n\} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Solved with forward substitution: $\{D\}^T = [1, -0.03333, -0.10009]$

AN EXAMPLE FOR MATRIX INVERSION

Step 2: A backward-substitution to compute $\{X\}$: $[U]\{X_1\} = \{D\}$

Back-substitution

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 1 \\ -0.03333 \\ -0.1009 \end{Bmatrix}$$

Solved with back substitution: $\{X_1\}^T = [0.33249, -0.00518, -0.01008]$

$$[A]^{-1} = \begin{bmatrix} 0.33249 & ? & ? \\ -0.00518 & ? & ? \\ -0.01008 & ? & ? \end{bmatrix}$$

AN EXAMPLE FOR MATRIX INVERSION

Step 3: Continue with A forward-substitution to compute {D} for second column: $[L] \{D\} = \{I_2\}$

Forward substitution ↓

$$\begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix}$$

Solved with forward substitution: $\{D\}^T$

Step 4: A backward-substitution to compute {X2}: $[U]\{X_2\} = \{D\}$

Back-substitution ↑

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \{D\}$$

Please
complete the
rest of the
calculation

FINAL ANSWER

$$[A]^{-1} = \begin{bmatrix} 0.3325 & 0.0049 & 0.0068 \\ -0.0052 & 0.1429 & 0.0042 \\ -0.0101 & 0.0027 & 0.0999 \end{bmatrix}$$